

Question 4

Consider the half-diamond symmetrical shape shown in Figure 3 (not to scale), with chord $c = 1\text{ m}$ and wedge angles of 4° , in a flow at Mach 2.5 (air, with $\gamma = 1.4$, $R = 287\text{ J kg}^{-1}\text{ K}^{-1}$; you may take the nearest values from the tables without interpolation).

Part (a)

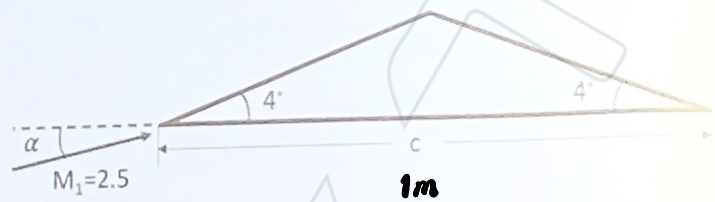
Sketch the expected flow structure for the following cases:

- Angle of attack $\alpha = 0^\circ$.
- Angle of attack $\alpha = 4^\circ$.
- Angle of attack $\alpha = 6^\circ$.

[5%]

[5%]

[5%]



Part (b)

Calculate the lift and drag coefficients for this section using shock-expansion theory when the angle of attack $\alpha = 6^\circ$.

[40%]

Part (c)

Calculate the lift and drag coefficients for this section at $\alpha = 6^\circ$ using Ackeret's linearised theory.

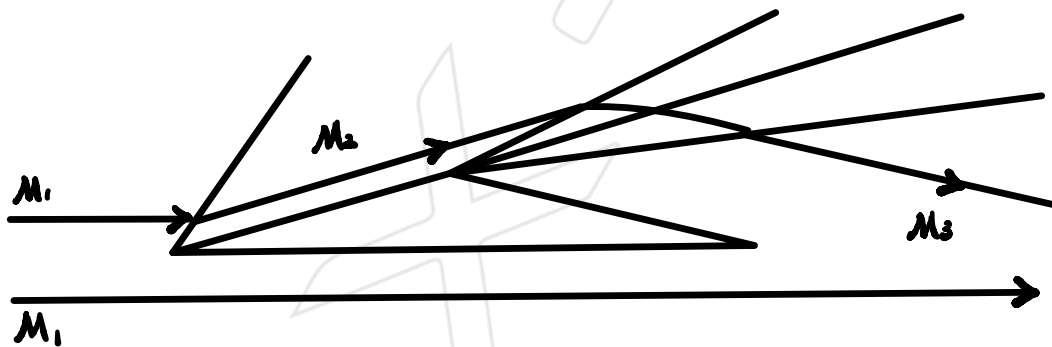
[40%]

Part (d)

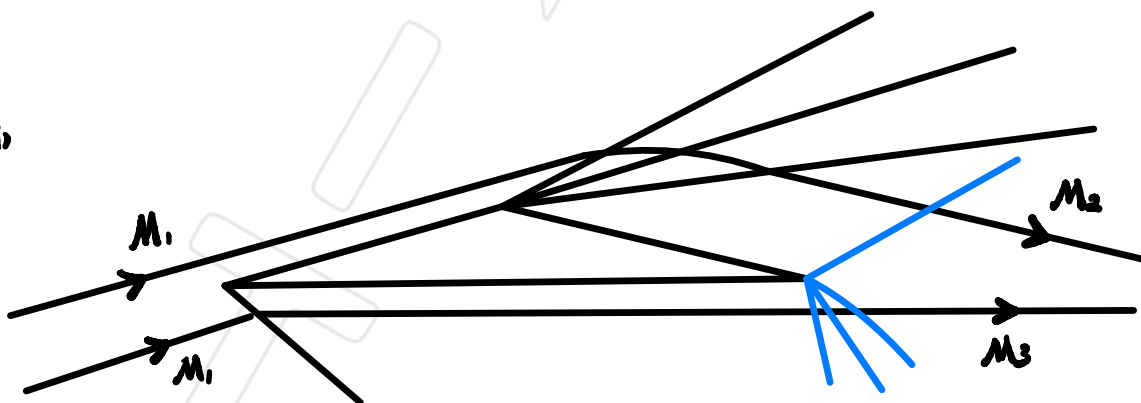
Calculate the error (in percentage) in the linearised method and briefly comment on your results.

[5%]

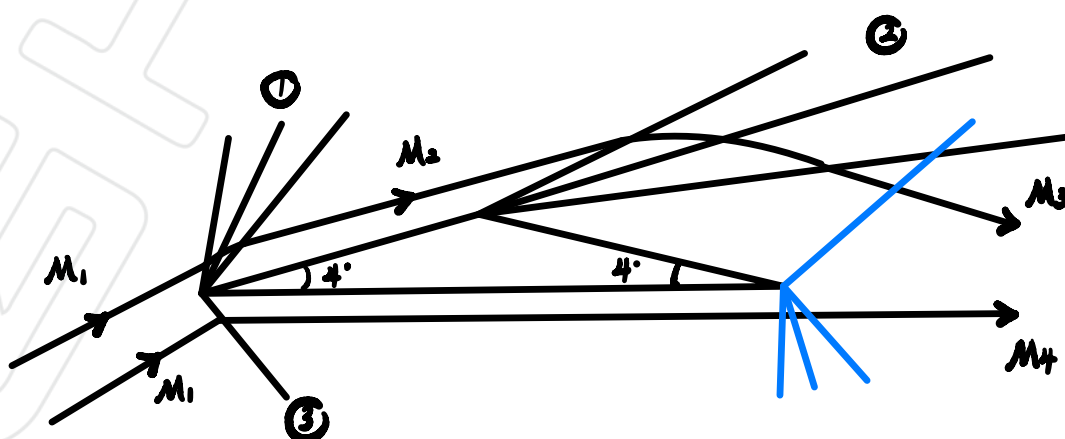
Q. i)



Q. ii)



Q. iii)



(b) When $\alpha = 6^\circ$, $M_1 = 2.5$, $\gamma(M_1) = 39.12^\circ$

$$\textcircled{1} \quad 6^\circ + \gamma(M_1) = 4^\circ + \gamma(M_2)$$

$$\therefore \gamma(M_2) = 2^\circ + \gamma(M_1) = 41.12^\circ$$

$$\therefore M_2 \approx 2.59$$

$$\therefore \frac{p_2}{p_1} = \frac{p_2}{p_{n2}} \frac{p_{n1}}{p_1} = \frac{17.086}{19.656} = 0.8692$$

$$\textcircled{2} \quad 4^\circ + \gamma(M_2) = -4^\circ + \gamma(M_3)$$

$$\therefore \gamma(M_3) = 8^\circ + \gamma(M_2) = 49.12^\circ$$

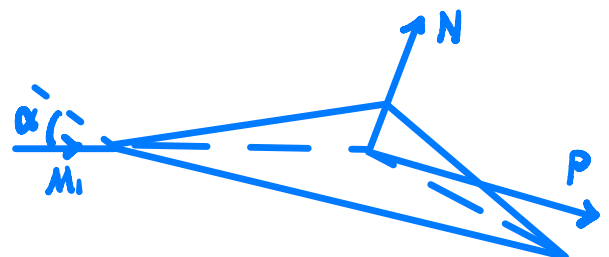
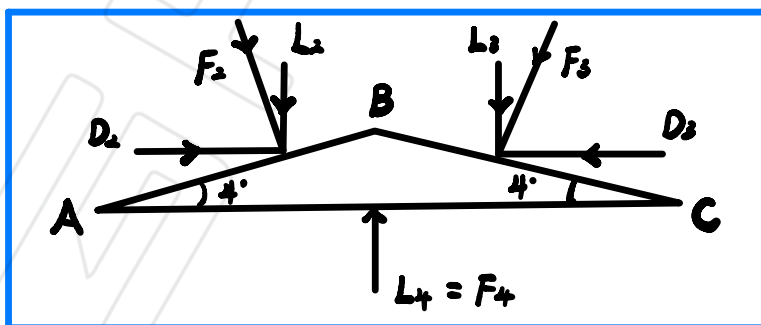
$$\therefore M_3 \approx 2.97$$

$$\therefore \frac{p_3}{p_1} = \frac{p_3}{p_{n3}} \frac{p_{n1}}{p_1} = \frac{17.086}{35.137} = 0.4863$$

$$\textcircled{3} \quad \theta = 6^\circ \quad \therefore \beta = 28.260^\circ \quad \therefore M_4 = 2.251$$

$$\therefore \frac{p_4}{p_1} = 1.468$$

$$\text{for 2-D. } AB = BC = \frac{\frac{1}{2}}{\cos 4^\circ} = 0.50122$$



$$\therefore D_2 = F_2 \sin 4^\circ = p_2 AB \sin 4^\circ = 0.03039 p_1$$

$$L_2 = F_2 \cos 4^\circ = p_2 AB \cos 4^\circ = 0.43460 p_1$$

$$D_3 = F_3 \sin 4^\circ = p_3 BC \sin 4^\circ = 0.01700 p_1$$

$$L_3 = F_3 \cos 4^\circ = p_3 BC \cos 4^\circ = 0.24315 p_1$$

$$L_4 = F_4 = p_4 AB = 1.468 p_1$$

$$\therefore D = D_2 - D_3 = 0.01399 p_1$$

$$L = L_4 - L_2 - L_3 = 0.79025 p_1$$

$$\therefore C_L = \frac{L}{\frac{1}{2} \rho_1 U_1^2} = \frac{2(0.79025 p_1)}{p_1 \cdot 8 M_1^2} = 0.1806 \approx 0.1793$$

$$C_D = \frac{D}{\frac{1}{2} \rho_1 U_1^2} = \frac{2(0.01399 p_1)}{p_1 \cdot 8 M_1^2} = 0.02192 \approx 1.977 \times 10^{-2}$$

$$(c) \quad y_c = \frac{1}{2} \tan(4^\circ) \left(\frac{x}{c}\right) \quad \text{for } 0 \leq x \leq \frac{1}{2}c$$

$$y_c = \frac{1}{4} \tan(4^\circ) - \frac{1}{2} \tan(4^\circ) \left(\frac{x - \frac{1}{2}c}{c}\right)$$

$$= \frac{1}{2} \tan(4^\circ) - \frac{1}{2} \tan(4^\circ) \left(\frac{x}{c}\right) \quad \text{for } \frac{1}{2}c \leq x \leq c$$

$$y_t = y_c \quad \text{in this case}$$

$\therefore$ By linear theory,

$$C_L = \frac{4\alpha}{\sqrt{M_1^2 - 1}} = 0.1828$$

$$N = (\Delta \bar{p}_c) \cdot c$$

$$= \frac{1}{2} (p_4 - p_3 + p_4 - p_2) \cdot c$$

$$= 0.79025 p_1$$

$$P = (\Delta \bar{p}_t) \cdot t$$

$$= 0.0133875 p_1$$

$$\therefore L = N \cdot \cos \alpha$$

$$- P \sin \alpha$$

$$= 0.7845 p_1$$

$$D = N \cdot \sin \alpha$$

$$+ P \cos \alpha$$

$$= 0.09592$$

$$\therefore \theta_c = \theta_t = \frac{1}{2} \tan(4^\circ) \text{ for } 0 \leq x \leq \frac{1}{2}c$$

$$= -\frac{1}{2} \tan(4^\circ) \text{ for } \frac{1}{2}c \leq x \leq c$$

$$\therefore C_D = \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\alpha^* + \int_0^1 (\theta_c^2 + \theta_t^2) d\left(\frac{x}{c}\right) \right)$$

$$= \frac{8\sqrt{21}}{21} \left(\alpha^* + \int_0^{\frac{1}{2}} \frac{1}{2} \tan^2(4^\circ) d\left(\frac{x}{c}\right) + \int_{\frac{1}{2}}^1 \frac{1}{2} \tan^2(4^\circ) d\left(\frac{x}{c}\right) \right)$$

$$= \frac{8\sqrt{21}}{21} \left(\alpha^* + \int_0^1 \frac{1}{2} \tan^2(4^\circ) d\left(\frac{x}{c}\right) \right)$$

$$= \frac{8\sqrt{21}}{21} \left(\alpha^* + \frac{1}{2} \tan^2(4^\circ) \right)$$

$$= 0.0234$$

(d) error for $C_L = 1.952\%$

error for $C_D = 6.752\%$

Linear approximation valid